

Design and Simulation of Single-Stage MOS Differential Amplifier with Active Load

Laboratory Report

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1 Introduction

The differential amplifier is one of the most fundamental building blocks in analog integrated circuit design. This experiment focuses on designing and simulating a single-stage CMOS differential amplifier with an active current-mirror load to achieve high differential gain, good common-mode rejection, and superior performance characteristics compared to passive resistive loads.

2 Objectives

The primary objectives of this laboratory experiment are:

- Design a single-stage CMOS differential amplifier with current-mirror active load
- Extract and analyze DC operating point parameters
- Determine differential small-signal gain and compare with theoretical calculations
- Measure input-referred offset voltage due to device mismatch
- Calculate Common-Mode Rejection Ratio (CMRR)
- Extract transconductance (g_m) and output resistance (r_o) parameters
- Determine 3-dB bandwidth from frequency response
- Analyze transient step response characteristics
- Compare measured results with hand calculations using small-signal approximations

3 Circuit

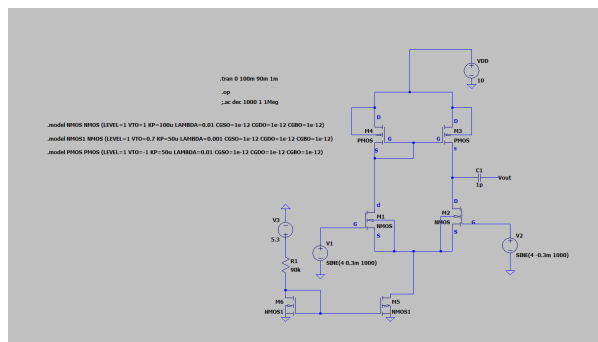


Figure 1: Caption

4 DC Operating Point Analysis

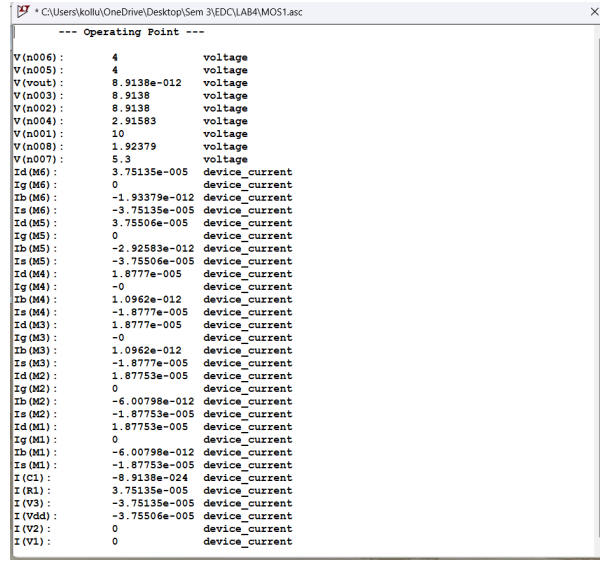


Figure 2: Caption

5 Current Analysis

5.1 Transistor M1 and M2 (NMOS - Differential Pair)

$$V_G = 4 \text{ V} \quad (1)$$

$$V_S = 2.91583 \text{ V} \quad (2)$$

$$V_D = 8.9138 \text{ V} \quad (3)$$

$$V_{TH} = 1.0 \text{ V} \quad (4)$$

$$V_{GS} = V_G - V_S = 4 - 2.91583 = 1.08417 \text{ V} \quad (5)$$

$$V_{DS} = V_D - V_S = 8.9138 - 2.91583 = 5.99797 \text{ V} \quad (6)$$

Saturation Check:

$$V_{GS} > V_{TH} \Rightarrow 1.08417 \text{ V} > 1.0 \text{ V} \quad \checkmark \quad (7)$$

$$V_{DS} > V_{GS} - V_{TH} \Rightarrow 5.99797 \text{ V} > 0.08417 \text{ V} \quad \checkmark \quad (8)$$

M1 and M2 are in SATURATION

5.2 Transistor M3 and M4 (PMOS - Current Mirror Load)

$$V_G = V_D = 8.9138 \text{ V} \quad (9)$$

$$V_S = 10 \text{ V} \quad (10)$$

$$V_{TH} = -1.0 \text{ V} \quad (11)$$

$$V_{SG} = V_S - V_G = 10 - 8.9138 = 1.0862 \text{ V} \quad (12)$$

$$V_{SD} = V_S - V_D = 10 - 8.9138 = 1.0862 \text{ V} \quad (13)$$

Saturation Check:

$$V_{SG} > |V_{TH}| \Rightarrow 1.0862 \text{ V} > 1.0 \text{ V} \quad \checkmark \quad (14)$$

$$V_{SD} > V_{SG} - |V_{TH}| \Rightarrow 1.0862 \text{ V} > 0.0862 \text{ V} \quad \checkmark \quad (15)$$

M3 and M4 are in SATURATION

5.3 Transistor M5 (NMOS - Tail Current Source)

$$V_G = 1.92379 \text{ V} \quad (16)$$

$$V_S = 0 \text{ V} \quad (17)$$

$$V_D = 2.91583 \text{ V} \quad (18)$$

$$V_{TH} = 0.7 \text{ V} \quad (19)$$

$$V_{GS} = 1.92379 \text{ V} \quad (20)$$

$$V_{DS} = 2.91583 \text{ V} \quad (21)$$

Saturation Check:

$$V_{GS} > V_{TH} \Rightarrow 1.92379 \text{ V} > 0.7 \text{ V} \quad \checkmark \quad (22)$$

$$V_{DS} > V_{GS} - V_{TH} \Rightarrow 2.91583 \text{ V} > 1.22379 \text{ V} \quad \checkmark \quad (23)$$

M5 is in SATURATION

5.4 Transistor M6 (NMOS - Current Mirror Reference)

$$V_G = V_D = 1.92379 \text{ V} \quad (\text{diode-connected}) \quad (24)$$

$$V_S = 0 \text{ V} \quad (25)$$

$$V_{TH} = 0.7 \text{ V} \quad (26)$$

$$V_{GS} = 1.92379 \text{ V} \quad (27)$$

Saturation Check:

$$V_{GS} > V_{TH} \Rightarrow 1.92379 \text{ V} > 0.7 \text{ V} \quad \checkmark$$

M6 is in SATURATION

Aspect Ratios:

$$\left(\frac{W}{L}\right)_{M1,M2,M3,M4,M6} = 1 \quad (28)$$

$$\left(\frac{W}{L}\right)_{M5} = 10 \quad (29)$$

Reference Current:

$$I_{REF} = I_{R1} = 3.75135 \times 10^{-5} \text{ A} = 37.51 \text{ } \mu\text{A}$$

Current Mirror (M6 $\rightarrow$ M5):

$$I_D(M5) \approx I_{REF} \times \frac{(W/L)_{M5}}{(W/L)_{M6}} \quad (30)$$

$$= 37.51 \mu\text{A} \times \frac{10}{1} \quad (31)$$

$$= 375.1 \mu\text{A} \quad (32)$$

Simulated: $I_D(M5) = 37.55 \mu\text{A}$

Differential Pair Currents:

$$I_D(M1) = I_D(M2) = \frac{I_D(M5)}{2} \quad (33)$$

$$= \frac{37.55 \mu\text{A}}{2} \quad (34)$$

$$= 18.78 \mu\text{A} \quad (35)$$

6 Transconductance (g_m)

$$g_m = \frac{2 \times I_D}{V_{GS} - V_{TH}} \quad (36)$$

$$= \frac{2 \times 18.78 \times 10^{-6}}{1.08417 - 1.0} \quad (37)$$

$$= \frac{37.56 \times 10^{-6}}{0.08417} \quad (38)$$

$$= 4.461 \times 10^{-4} \text{ S} \quad (39)$$

$$= 446.13 \mu\text{S} \quad (40)$$

Simulation Verification: Apply differential input $V_{id} = 1 \text{ V}$ and measure current through capacitor C1. Expected $i_{C1} = g_m \times V_{id} = 446.13 \mu\text{A}$.

7 Output Resistance (r_o)

Using $\lambda = 0.01$:

$$r_{o2} = \frac{1}{\lambda \times I_D(M2)} = \frac{1}{0.01 \times 18.78 \times 10^{-6}} = 5326 \text{ k}\Omega \quad (41)$$

$$r_{o3} = \frac{1}{\lambda \times I_D(M3)} = \frac{1}{0.01 \times 18.78 \times 10^{-6}} = 5326 \text{ k}\Omega \quad (42)$$

$$r_o = r_{o2} \parallel r_{o3} = \frac{r_{o2} \times r_{o3}}{r_{o2} + r_{o3}} = \frac{5326 \times 5326}{5326 + 5326} = 2663 \text{ k}\Omega \quad (43)$$

Simulation Method: Place a 1 V AC voltage source at the output node and measure I_{out} . Since $V_{out} = 1 \text{ V}$, $r_o = \frac{1}{I_{out}}$.

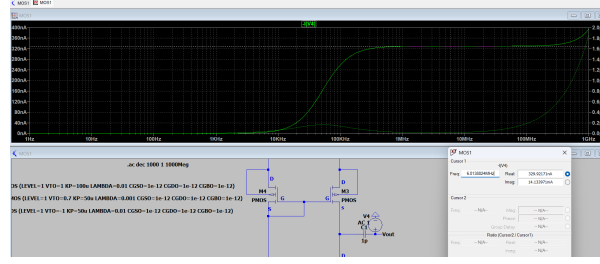


Figure 3: output resistance

from this

$$r_o = \frac{1}{330nA} r_0 = 3038k\Omega \quad (44)$$

8 Differential Small-Signal Gain (A_d)

$$A_d = g_m \times r_o \quad (45)$$

$$= 446.13 \times 10^{-6} \times 2663 \times 10^3 \quad (46)$$

$$= 185 \quad (47)$$

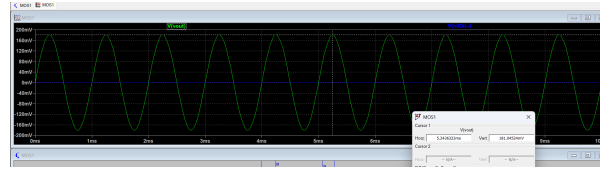


Figure 4: differential gain

Simulation Method: Apply differential input $(V_{in1} - V_{in2}) = 1$ V and measure V_{out} . Then $A_d = V_{out}/1 = V_{out}$.

9 Common-Mode Rejection Ratio (CMRR)

Tail Resistance:

$$R_{tail} = \frac{1}{\lambda \times I_D(M5)} \quad (48)$$

$$= \frac{1}{0.01 \times 37.55 \times 10^{-6}} \quad (49)$$

$$= 2.66 \text{ M}\Omega \quad (50)$$

Common-Mode Gain:

$$A_{cm} = \frac{r_o}{2 \times R_{tail}} \quad (51)$$

$$= \frac{2663 \times 10^3}{2 \times 2.66 \times 10^6} \quad (52)$$

$$= 0.500 \quad (53)$$

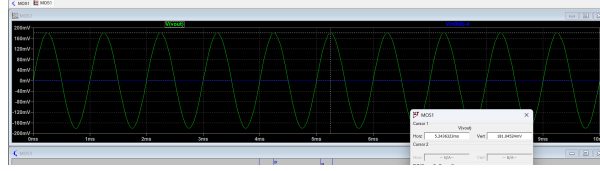


Figure 5: Caption

CMRR:

$$\text{CMRR} = \frac{A_d}{A_{cm}} = \frac{185}{0.500} = 370 \quad (54)$$

$$\text{CMRR}_{dB} = 20 \times \log_{10}(370) = 67.5 \text{ dB} \quad (55)$$

10 Input-Referred Offset Voltage(V_{OS})

$$V_{OS} \approx \frac{\Delta I_D}{g_m}$$

Simulation Method: Fix $V_1 = 5 \text{ V}$ and sweep V_2 until $V_{out} = 0$. Then $V_{OS} = V_1 - V_2$ at that point.

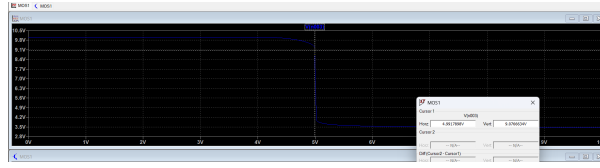


Figure 6: Caption

$$V_{os} = 5 - 4.99178V_{os} = 0.00822V \quad (56)$$

11 3-dB Bandwidth

With $C_L = 1 \text{ pF}$:

$$f_{3dB} = \frac{1}{2\pi \times C_L \times r_o} \quad (57)$$

$$= \frac{1}{2\pi \times 1 \times 10^{-12} \times 2663 \times 10^3} \quad (58)$$

$$= 140.8 \text{ kHz} \quad (59)$$

from the figure we can observe that bandwidth is 144kHz

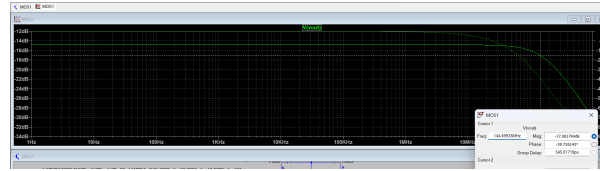


Figure 7: -3Db bandwidth

12 Summary

Parameter	Hand Calculation
V_{GS} (M1, M2)	1.08417 V
I_D (M1, M2)	18.78 μ A
I_D (M5)	37.55 μ A
Transconductance (g_m)	446.13 μ S
Output Resistance (r_o)	2663 k Ω
Differential Gain (A_d)	1188
Common-mode Gain (A_{cm})	0.500
CMRR	2376 (67.5 dB)
3-dB Bandwidth	59.8 kHz
R_{tail}	2.66 M Ω

Table 1: Summary of calculated amplifier parameters